Niramaya Health AI - Viva Voce & Technical Defense Guide

Comprehensive Master Question-Answer Companion for Final Year Project Examination

Project: Niramaya AI Symptoms Checker | Domain: Healthcare AI & Biomedical NLP | Stack: Python, Flask, Scikit-Learn, Transformers, PostgreSQL/SQLite | Accuracy: 91% (0.91)

1. Machine Learning, AI Models & Evaluation Metrics

QUESTION 1

Aapke project ki exact accuracy kitni hai aur yeh kaise evaluate ki gayi?

Hamare **Logistic Regression (LRv1 & LRv2)** disease prediction model ki overall **Accuracy 91% (0.91)** hai.

Metric Name	Calculated Score	Clinical Significance in Healthcare
Overall Accuracy	**91% (0.91)**	Overall correct classification across all 40+ disease categories.
Precision	**89% (0.89)**	False Positives minimize karta hai (unnecessary panic prevent karta hai).
Recall (Sensitivity)	**91% (0.91)**	False Negatives minimize karta hai (actual disease miss nahi honi chahiye).
F1-Score	**89% (0.89)**	Precision aur Recall ka balanced harmonic mean.
5-Fold Cross Validation	**89% (0.89)**	Verifies model stability and absence of data leakage/overfitting.

Evaluation Methodology: Model ko Kaggle ke *Symptoms-to-Disease* dataset par train kiya gaya (80% Training set, 20% Testing set split). Scikit-Learn ke `classification_report`, `confusion_matrix`, aur `cross_val_score` ke through mathematically test kiya gaya hai. Iska Confusion Matrix website ke *Our Models* section me verified hai.

> **Examiner Pro-Tip:** Speak confidently: *"Sir, in clinical diagnostics, Recall (91%) is considered more critical than raw accuracy because missing a true disease (False Negative) can be life-threatening."*

QUESTION 2

Why did you use Logistic Regression? Baki kafi regression/models hain, to yahi kyu use kiya?

Key Clarification: Humne **Logistic Regression** use kiya hai, **Linear Regression nahi**. Linear Regression continuous values predict karta hai (e.g. house price), jabki Disease Prediction ek **Discrete Multi-Class Classification Problem** hai.

Baki complex models (Random Forest, SVM, Deep Neural Networks) ke mukable Logistic Regression chunne ke **4 technical reasons** hain:

- **High Dimensional Sparse Binary Vectors:** Dataset me 130+ symptoms binary features hain (1 = present, 0 = absent). Jab user 3-4 symptoms deta hai, to 95% vector zeros hota hai. Linear boundary sparse vectors par overfit nahi hoti aur best convergence deti hai.
- **Probability Estimation via Softmax:** Logistic Regression mathematically classes ki probabilities calculate karti hai. Isse hum user ko prediction ke saath *Confidence Score* provide kar sakte hain.
- **Resistance to Overfitting:** Deep Neural Networks aur large Random Forests tabular discrete datasets par training data ko memorize kar lete hain (overfitting). Logistic Regression generalizes exceptionally well.
- **Lightweight & Instant Inference:** Model file (`.pkl`) sirf ~150KB ki hai. Inference 2-5 milliseconds me execute ho jaata hai, jisse bina GPU ke normal laptop par 100% offline run ho sakta hai.

QUESTION 3

Gemini aur BioGPT kya hain? Aur dono me technical difference kya hai?

Dono **Large Language Models (LLMs)** hain, lekin unki training aur specialization alag hai:

- **BioGPT (Microsoft Research):** Yeh ek *Domain-Specific Transformer* model hai jise PubMed aur biomedical literature ke 15+ million clinical research abstracts par pre-train kiya gaya hai. Iska biomedical vocabulary understanding (disease terms, medical symptoms, clinical relations) unmatched hai.
- **Gemini (Google DeepMind):** Yeh ek *General Multimodal Foundation LLM* hai jo complex, multi-paragraph conversational advice, home remedies, lifestyle suggestions, aur precautions generate karne me superior hai.

Project me role: User specialized medical text understanding chahta hai to BioGPT choose karta hai; jab conversational, structured care advice chahiye to Gemini choose karta hai.

QUESTION 4

Hugging Face kya hai aur aapke project me iska use kaha kiya gaya hai?

Hugging Face AI aur Open-Source Machine Learning ka sabse bada ecosystem hai (often called 'GitHub for AI Models').

Niramaya me use: Hamare backend `backend/routes/ask_AI_API.py` me Hugging Face ki `transformers` library use ki gayi hai:

- `BioGptTokenizer` : User ke medical text ko mathematical token embeddings me transform karta hai.
- `BioGptForCausalLM` : Microsoft ke pre-trained weights (`microsoft/biogpt`) load karta hai.
- `pipeline('text-generation')` : Real-time clinical text generation handle karta hai.

QUESTION 5

LRv1 aur LRv2 me kya antar (difference) hai?

- **LRv1 (Baseline Clinical Awareness):** User ke input symptoms ko match karke multi-class classification se probable disease predict karta hai aur `Symptom_Description.csv` se verified clinical overview deta hai.
- **LRv2 (Actionable Preventive Care):** Same high-accuracy Logistic Regression model use karta hai, lekin output me disease prediction aur description ke saath-saath **Tailored 4-Step Precautions** (`Symptom_Precaution.csv`) provide karta hai (e.g. dietary adjustments, lifestyle precautions, urgent consultation warnings).

2. System Architecture & Data Flow

QUESTION 6

Inme answers API ke through aate hain ya Database me pehle se stored hain?

Yeh ek **Hybrid Intelligent Architecture** hai. Answers database me hardcoded nahi hote, balki real-time compute hote hain:

- **ML Inference (LRv1/LRv2):** Jab user query karta hai, Python backend `lr_model.predict(input_vector)` run karta hai. Yeh calculation real-time CPU par hoti hai.
- **LLM Inference (BioGPT / Gemini):** BioGPT local Hugging Face transformer model se aur Gemini Google secure API endpoint se dynamically text generate karta hai.
- **Database ka Role:** Database (`PostgreSQL / SQLite`) ka kaam answers generate karna nahi hai; balki har consultation ka *Audit Log* store karna hai (`account_id` , `member_id` , `user_input` , `bot_response` , `timestamp`) taaki user aur family members apni historical health trends dekh sakein.

Examiner Tip: Agar examiner puche: *"Python use kiya hai isliye data waha se aa raha hai?"* Answer: *"Yes Sir, Python Flask backend me controller aur inference engine hai, jo models run karke dynamically JSON response frontend ko deliver karta hai."*

QUESTION 7

Step-by-step user input se output tak ka complete Data Flow explain karo.

1. **Frontend:** User web interface (`question_input.html`) par symptoms likhta hai aur model select karta hai.
2. **HTTP Request:** JavaScript asynchronous `fetch()` call trigger karta hai `POST /asklrv1` ya `POST /askgemini` par with JSON payload.
3. **Flask Routing:** Flask ka modular Blueprint request intercept karta hai.
4. **Preprocessing & Normalization:** Regex engine (`parse_symptoms()`) user ke comma/space separated text ko dataset ke standardized 130+ symptom vocabulary me map karta hai.
5. **Inference:** Feature vector generate hota hai aur AI model prediction compute karta hai.
6. **Database Logging:** Consultation securely `chat_history_tbl` me insert hoti hai with family member association.
7. **HTTP Response:** Backend JSON response return karta hai, aur frontend DOM dynamically cards render karta hai.

3. Database Design & Scalability

QUESTION 8

Database schema explain karo. Accounts aur Members ka relation kaisa hai?

Database me **One-to-Many Relational Design** follow kiya gaya hai:

- `account_tbl`: Family / Primary User ka account (Email, Password Hash, Family Name).
- `member_tbl`: Family ke members (Grandfather, Child, Self) with age, gender, height, weight. Linked via `FOREIGN KEY (account_id) REFERENCES account_tbl(account_id) ON DELETE CASCADE`.
- `chat_history_tbl`: Har member ki individual health queries, AI model used, aur predictions.
- `admin_tbl` & `admin_tokens_tbl`: Administrative privilege levels aur token session security.
- `health_content_tbl`: Curated wellness articles aur preventive advice.

Advantage: Ek hi email par pure parivaar ke alag-alag members ka health track maintain hota hai bina unke health records mix hue.

QUESTION 9

Dual Database Support (PostgreSQL + SQLite Fallback) kyu aur kaise implement kiya?

Enterprise reliability aur zero-dependency portability ke liye humne dual database engine wrapper banaya hai:

- **PostgreSQL:** Production environment me high-concurrency ACID transactions ke liye use hota hai.
- **SQLite Seamless Fallback:** Agar student laptop par PostgreSQL service install nahi hai ya cloud container me dedicated DB offline hai, to backend connection timeout hote hi bina crash hue lightweight local `niramaya.db` me shift ho jaata hai.
- **Custom SQLiteConnWrapper:** Humne wrapper class banayi hai jo PostgreSQL ke dialect (`%s`, `ILIKE`, `RETURNING`) ko runtime par translate karti hai taaki code dono engines par identical chale.

4. Security, Authentication & Deployment

QUESTION 10

User aur Admin passwords kaise secure kiye gaye hain?

Hamara application strict **Zero Plaintext Password** rule follow karta hai:

- **Bcrypt Adaptive Hashing:** Passwords ko `bcrypt.hashpw(password, bcrypt.gensalt(12))` ke through cryptographic salt ke sath hash kiya jata hai.
- **Rainbow Table Protection:** Salt randomized hota hai, jisse dictionary attacks aur precomputed rainbow tables fail ho jaate hain.
- **Constant-Time Verification:** Verification `bcrypt.checkpw()` se hoti hai, jo timing attacks prevent karti hai.
- **Admin Token Expiry:** Admin session me cryptographically secure hex tokens generate hote hain jo 24 hours baad automatically expire aur invalidate ho jaate hain.

QUESTION 11

Render par deploy karte waqt Gunicorn kyu use kiya? Flask ka internal server kyu nahi?

- Flask ka default `app.run()` ek **Development Server** hai jo single-threaded hota hai aur production traffic, connection pooling, ya worker timeouts handle nahi kar sakta.
- **Gunicorn (Green Unicorn)** ek industry-standard **WSGI (Web Server Gateway Interface) HTTP Server** hai. Yeh pre-fork worker model use karta hai jo memory leaks handle karta hai, incoming requests ko manage karta hai, aur process crash hone par worker ko instantly revive karta hai.

5. Healthcare Ethics, AI Hallucination & Disclaimers

QUESTION 12

Agar AI galat bimari predict kar de to patient ki safety kaise ensure hoti hai?

Healthcare AI me **Patient Safety First** principle follow kiya gaya hai:

- **Non-Diagnostic Scope:** Niramaya ek *Preliminary Symptom Awareness System* hai, Certified Doctor ka replacement nahi.
- **Mandatory Disclaimers:** Har single AI response ke sath mandatory medical disclaimer display hota hai: *"Niramaya provides informational insights only. Please consult a licensed medical professional before taking any medication."*
- **Emergency Escalation:** Severe symptoms ke case me UI user ko emergency helpline aur physical clinical checkup consult karne ka alert deti hai.
- **Hallucination Prevention:** LRv1/LRv2 open-ended answers hallucinate nahi kar sakte kyunki wo strictly clinical datasets par constrained hain.

6. Future Scope & Potential Improvements

QUESTION 13

Aage chal kar is project me aur kya enhance kiya ja sakta hai?

- **Medical Report & Image Uploads (Multimodal AI):** X-rays, blood reports, aur prescription images analyze karne ke liye Vision Transformer (ViT) ya CNN integrate karna.
- **Voice-to-Symptom Input (Speech-to-Text):** Regional languages me bol kar symptoms batane ki suvidha for rural healthcare accessibility.
- **Teleconsultation Integration:** Predicted disease ke basis par verified local doctors ke sath direct video consultation booking.
- **Continuous Model Retraining (MLOps):** Anonymized clinical feedback ke sath model weights ko regularly fine-tune karna.

© 2026 Niramaya Health AI Symptoms Checker Project — Confidential Academic Project Defense Guide